QRICS CALIBRATION

(Mechanical, Electro-Technical & Thermal)

CC-2747

CALIBRATION CERTIFICATE

Certificate No.	Issue Date	ULR No	Calibration Date	Next Due Date	No. of Pages
QC/23-24/12271	05/12/2023	CC274723000017088F	04/12/2023	03/12/2025	1 of 1

MRROA No. : QC1242 **Date of Receipt** 04/12/2023 **Customer DC. No.** : OFFICE003/23-24 dt: 04/12/2023

Customer Address : M/s Nihva Technologies Pvt. Ltd., 187/G-Block, 1st Floor, Thermax Chowk, Chinchwad, Pune-411019

Calibration Location : In Lab
Condition of Item : OK
UUC* Details : Coat Thickness Gauge
Range : 0-1350 µm
Identification No. : DFT-01
Id No. taken from : As per challan
Standard Referred : IS 6012

Make : Mextech
L.C. : 0.1 µm / 1 µm
Sl. No. : N1061412
Accuracy : ± 1.5% of reading & 2 µm whichever is more

Calibration Method : QS/CPM/21

Test Purpose : To check the actual coat thickness gauge with pre calibrated Qrics foils.

Environmental Condition : 20 ± 1° C, Humidity 40 – 60%

Standards / Equipment used for calibration:			Traceable to National / International standards through unbreakable chain of NABL accredited calibration laboratory.		
Sr. No.	Standard / Equipment	Id. No.	Cert. No.	NABL Cert. No.	Validity
1.	Foils	QS/MM/09 (A to G)	QC/22-23/10851 dated 02/01/2023	CC-2747	01/01/2024

Observations : **Pre Calibration Status** : Worth calibrating
: **Post Calibration Status** : As follows reading average value of three reading,

For Ferrous

Sl. No.	Actual Foils Size µm	CTG Set to Actual Foils size.	Actual Reading on CTG in µm	Repeatability in µm	Error	Error in %
1	25	25	24.6	0.2	-0.4	-1.6
2	51	51	51.4	1.0	0.4	0.8
3	125	125	125.0	3.0	0.0	0.0
4	252	252	252.0	1.0	0.0	0.0
5	515	515	516.0	4.0	1.0	0.2
6	708	708	709.0	2.0	1.0	0.1

For Non- Ferrous

Sl. No.	Actual Foils Size µm	CTG Set to Actual Foils size.	Actual Reading on CTG in µm	Repeatability in µm	Error	Error in %
1	25	25	25.2	0.8	0.2	0.8
2	51	51	51.1	0.9	0.1	0.2
3	125	125	124.0	2.0	-1.0	-0.8
4	252	252	253.0	2.0	1.0	0.4
5	515	515	515.0	4.0	0.0	0.0
6	708	708	710.0	5.0	2.0	0.3

Vijay V Satpute QUEST 3046 Pune

Note: The customer is advised to select the shim nearest to the value under measurement and set to the true value of shim & zero by using ± buttons on meter. Then measure the actual value on component.

Uncertainty of Measurement : ± 4.74 µm

(The reported uncertainty at above reading is the expanded uncertainty of measurement obtained by multiplying the standard uncertainty by the coverage factor k=2 corresponds to a confidence level of 95.45 %.)

*** END ***

Reviewed 08/12/23

Calibrated By N. M. Shinde, Technical Assistant

Approved By Ankita Bhuvad, Authorized Signatory

User discretion:

Note: 1) Any error / Complaint must be resolved with lab within 3 days from the receipt of certificate in written.

Regd. Office & Labs : "Paras Complex", GP/67, Office No. 2 & 7 to 10 (Basement), Thermax Chowk, Chinchwad, Pune - 411 019.
Customer Care : 020-27371831, 07276710888 & Accounts : 020-27370204, Suggestion, Feedback & Complaint : 9822069138, E-mail : qrics@hotmail.com, web : www.qrics.in

QRICS CALIBRATION

(Mechanical, Electro-Technical & Thermal)

CC-2747

CALIBRATION CERTIFICATE

Certificate No.	Issue Date	ULR No	Calibration Date	Next Due Date	No. of Pages
QC/23-24/12270	05/12/2023	CC274723000017087F	04/12/2023	03/12/2025	1 of 1

MRROA No. : QC1242 **Date of Receipt** 04/12/2023 **Customer DC. No.** : OFFICE003/23-24 dt: 04/12/2023

Customer Address : M/s Nihva Technologies Pvt. Ltd.,
187/G-Block, 1st Floor, Thermax Chowk, Chinchwad, Pune-411019

Calibration Location : In Lab
Condition of Item : OK
UUC* Details : Foils
Size of foils : 125 to 705 µm
Id. No. of foils : Nil
Id. No. taken from : NA
Calibration Method : QS/CPM/21

Make : Unknown
L.C. : NA
Sl. No. : As below

Test Purpose : To check the actual thickness of foils

Environmental Condition : 20 ± 1° C, Humidity 40 – 60%

Standards / Equipment used for calibration:			Traceable to National / International standards through unbreakable chain of NABL accredited calibration laboratory.		
Sr. No.	Standard / Equipment	Id. No.	Cert. No.	NABL Cert. No.	Validity
1.	Universal Length measuring machine LMM 300T	QS/MM/37	CGIL/2023/1125/I-001 dt: 25/11/2023	CC-2656	24/11/2025

Observations : **Pre Calibration Status** : Worth calibrating,
: **Post Calibration Status** : As Follows

Mechanical Calibration: Dimension (Basic Measuring Instruments, Gauges etc.)

Sl. No.	Sl. No.of Foils	Specified Foils Size in µm	Actual Foils Size in µm
1	CM 100	125 ± 1	126
2	CM 300	282 ± 3	283
3	CM 500	393 ± 5	389
4	CM 800	705 ± 8	707

Uncertainty of Measurement : ± 2.2 µm

(The reported uncertainty at above reading is the expanded uncertainty of measurement obtained by multiplying the standard uncertainty by the coverage factor k=2 corresponds to a confidence level of 95.45%.)

Note: *UUC =Unit Under Calibration

Reviewed
08/12/23

*** END ***

Vijay V Satpute QUEST 3046 Pune

22/3/24

Calibrated By N. M. Shinde, Technical Assistant

Approved By Ankita Bhuvad, Authorized Signatory

User discretion:

Note: 1) Any error / Complaint must be resolved with lab within 3 days from the receipt of certificate in written.

Regd. Office & Labs : "Paras Complex", GP/67, Office No. 2 & 7 to 10 (Basement), Thermax Chowk, Chinchwad, Pune - 411 019.
Customer Care : 020-27371831, 07276710888 & Accounts : 020-27370204, Suggestion, Feedback & Complaint : 9822069138, E-mail : qrics@hotmail.com, web : www.qrics.in

P.T.O.

CERTIFICATE OF INSPECTION / 検査成績書

発行No. / Issue No. : 4501357404

Product name / 品名	Digimatic Caliper / ﾃﾞｼﾞﾏﾁｯｸｷｬﾘﾊﾟ
Model / 符号	CD-P12"S
Code No. / コードNo.	500-754-20
Serial No. / 製造No.	0057320
Measuring range / 測定範囲	0-300mm / 0-12in
Resolution / 最小表示量	0.01mm / 0.0005in
Reference temperature / 標準温度	20℃
QC Manager T.Shinohara	T. Shinohara

Inspection standard : Mitutoyo standard

Based on : ISO13385-1 (ASME B89)

Reference standard for inspection is traceable to NMIJ AIST through JCSS which is equivalent to NIST and PTB traceability by CIPM MRA.

Reviewed

この検査成績書には安心してご利用いただけますよう出荷時の検査データを記載していますが、校正証明書取得用にはご利用できません。

Mitutoyo Corporation

(1) Result of inspection / 検査結果

Measured length 測定長		Maximum permissible error 最大許容誤差(MPE)	Error of indication / 指示誤差	
			Outside / 外側	Inside / 内側
Unit : mm 単位 : mm	φ5	±0.04	---	-0.04
	0	±0.02	0.00	---
	50		+0.01	-0.01
	100		+0.01	-0.02
	150		+0.01	-0.02
	200		+0.01	-0.02
	300	±0.03	+0.01	-0.01
Unit : in 単位 : in	φ0.2	±0.0020	---	-0.0020
	0	±0.0010	0.0000	---
	2		+0.0005	-0.0005
	4		+0.0005	-0.0010
	6		+0.0005	-0.0010
	8		+0.0005	-0.0010
	12	±0.0015	+0.0005	-0.0005

(2) Judgment / 判定 : Passed / 合格

U0049-00223(10)

Vijay V Satpute QUEST 3046 Pune

22/3/22

CERTIFICATE OF INSPECTION / 検査成績書

Issue No./発行No. 22L1902BC01

Product name/品名	Digimatic Micrometer
Model/符号	MDC-25PX
Code No./コードNo.	293-240-30
Serial No./製造No.	72942156

Measuring range/測定範囲	0-25mm
Resolution/最小表示量	0.001mm
Reference temperature/標準温度	20°C
QC Manager T.Komori	

1.Result of inspection/検査結果

Inspection standard : Mitutoyo standard

1) Flatness of measuring face/平面度

	Permissible value /許容値[μm]	Measured value /実測値[μm]
Anvil/アンビル	0.3	0.0
Spindle/スピンドル	0.3	0.0

2) Parallelism of measuring faces/平行度

Permissible value /許容値[μm]	Measured value /実測値[μm]
1.0	0.0

3) Full surface contact error J/全測定面接触誤差

Measured length /測定長[mm]	Maximum permissible error /最大許容誤差[μm]	Error of indication /指示誤差[μm]
0.00	±1	0
7.70		0
12.90		−1
17.60		0
22.80		0
25.00		0

2.Judgment/判定 : Passed/合格

この検査成績書には安心してご利用いただけますよう出荷時の検査データを記載していますが、校正証明書取得用にはご利用できません。

Based on : JISB7502(2016),ISO3611(2010),DIN863-1(2017)

Reference standard for inspection is traceable to NMIJ AIST through JCSS which is equivalent to NIST and PTB traceability by CIPM MRA.

Reviewed

Mitutoyo Corporation

Vijay V Satpute
QUEST
3046
Pune

QRICS CALIBRATION

(Mechanical, Electro-Technical & Thermal)

CC-2747

CALIBRATION CERTIFICATE

Certificate No.	Issue Date	ULR No	Calibration Date	Next Due Date	No. of Pages
QC/23-24/12377	09/12/2023	CC274723000017184F	09/12/2023	08/12/2024	1 of 2

MRROA No. : QC1267 **Date of Receipt** 09/12/2023 **Customer DC No.** : OFFICE004/23-24 Dt: 09/12/2023

Customer : M/s Nihva Technologies Pvt. Ltd.,

Address 187/G-Block, 1st Floor, Thermax Chowk, Chinchwad, Pune-411019

Calibration Location	: In Lab		
Condition of Item	: OK		
UUC* Details	: Steel Tape (Trapezoid Type Ring)	**Make**	: Freemans
Range	: 15 mtrs	**L.C.**	: 1 mm
Identification No.	: NTPL/ST/08	**Sl. No.**	: Nil
Id No. taken from	: As per challan	**Acceptance Criteria**	: ± 2 mm (As per customer)
Standard Referred	: IS 1269		

Test Purpose : To check the error in accordance to the standard (Calibration Method : QS/CPM/66)

Environmental Condition : 20 ± 1° C, Humidity 40 – 60%

Observations **: Pre Calibration Status** : Worth Calibrating,

Standards / Equipment used for calibration:			Traceable to National / International standards through unbreakable chain of NABL accredited calibration laboratory.		
Sr. No	**Standard / Equipment**	**Id. No.**	**Cert. No.**	**NABL Cert. No.**	**Validity**
1.	Tape & Scale Unit	QS/ME/75	SCL/23-24/11/204-001 dt 18/11/2023	CC-2282	17/11/2024

Note: *UUC =Unit Under Calibration

Vijay V Satpute QUEST 3046 Pune

Calibrated By Ankita Bhuvad (Technical Assistant)

QRICS PUNE CALIBRATION

Approved By N. M. Shinde Technical Manager

Regd. Office & Labs : "Paras Complex", GP/67, Office No. 2 & 7 to 10 (Basement), Thermax Chowk, Chinchwad, Pune - 411 019.
Customer Care : 020-27371831, 07276710888 & Accounts : 020-27370204, Suggestion, Feedback & Complaint : 9822069138, E-mail : qrics@hotmail.com, web : www.qrics.in

P.T.O.

QRICS CALIBRATION

(Mechanical, Electro-Technical & Thermal)

CC-2747

CALIBRATION CERTIFICATE

Certificate No.	Issue Date	ULR No	Calibration Date	Next Due Date	No. of Pages
QC/23-24/12377	09/12/2023	CC274723000017184F	09/12/2023	08/12/2024	2 of 2

Observations : **Post Calibration Status** : Since this tape is not with 'zero away from trapezoid ring' as per standard following readings are taken from the outer face of trapezoid ring

Mechanical Calibration: Dimension (Basic-Measuring Instrument, Gauges etc.)

Parameter distance between outer face of ring (in mtr.)	Readings taken at master values (in mm)	Values of UUC (in mm)	Error (in mm)
0-0.1	99.990	100.495	0.505
0-1.0	999.955	1000.315	0.360
0-2.0	1999.955	2000.245	0.290
0-3.0	2999.955	3000.405	0.450
0-4.0	3999.955	4000.270	0.315
0-5.0	4999.955	5000.030	0.075
0-6.0	5999.955	5999.960	0.005
0-7.0	6999.955	7000.220	0.265
0-8.0	7999.955	7999.935	-0.020
0-9.0	8999.955	8999.575	-0.380
0-10.0	9999.955	9999.500	-0.455
0-11.0	10999.955	10999.115	-0.840
0-12.0	11999.955	11998.845	-1.110
0-13.0	12999.955	12998.705	-1.250
0-14.0	13999.955	13998.420	-1.535
0-15.0	14999.955	14998.580	-1.375

NOTE

Maximum permissible error for total length in steel tape measure as per IS 1269(Part 2) is as below.(where 'L' is in meter)

Class I	Class II	Class III
± (0.1+0.1 x L)	± (0.3+0.2 x L)	± (0.6+0.4 x L)

1. For steel tapes of length upto 5 mtr. Permissible values are as per class I & II only.
2. For steel tapes of length above 5 mtr. Permissible values are as per class I, II and III.
3. For tapes in service the permissible value is equal to TWICE the values indicated in the above table.

Uncertainty of Measurement : ± 980 µm

(The reported uncertainty at above reading is the expanded uncertainty of measurement obtained by multiplying the standard uncertainty by the coverage factor k=2 corresponds to a confidence level of 95.45%.)

Reviewed 14/12/23

Vijay V Satpute QUEST 3046 Pune

*** END ***

QRICS PUNE CALIBRATION

Calibrated By Ankita Bhuvad (Technical Assistant)

Approved By N. M. Shinde Technical Manager

User discretion:

Note: 1) Any error / Complaint must be resolved with lab within 3 days from the receipt of certificate in written.

Regd. Office & Labs : "Paras Complex", GP/67, Office No. 2 & 7 to 10 (Basement), Thermax Chowk, Chinchwad, Pune - 411 019.
Customer Care : 020-27371831, 07276710888 & Accounts : 020-27370204, Suggestion, Feedback & Complaint : 9822069138, E-mail : qrics@hotmail.com, web : www.qrics.in

P.T.O.

QRICS CALIBRATION

(Mechanical, Electro-Technical & Thermal)

CC-2747

CALIBRATION CERTIFICATE

Certificate No.	Issue Date	ULR No	Calibration Date	Next Due Date	No. of Pages
QC/23-24/12379	09/12/2023	CC274723000017186F	09/12/2023	08/12/2024	1 of 1

MRROA No. : QC1267 **Date of Receipt** 09/12/2023 **Customer DC No.** : OFFICE004/23-24 Dt: 09/12/2023

Customer : M/s Nihva Technologies Pvt. Ltd.,
Address 187/G-Block, 1st Floor, Thermax Chowk, Chinchwad, Pune-411019

Calibration Location : In Lab
Condition of Item : OK
UUC* Details : Steel Tape (L Type) **Make** : Freemans
Range : 5 mtrs **L.C.** : 1 mm
Identification No. : NTPL/ST/07 **Sl. No.** : Nil
Id No. taken from : As per challan **Acceptance Criteria** : ± 2 mm (As per customer)
Standard Referred : IS 1269
Test Purpose : To check the error in accordance to the standard (Calibration Method : QS/CPM/66)
Environmental Condition : 20 ± 1° C, Humidity 40 – 60%

Standards / Equipment used for calibration:			Traceable to National / International standards through unbreakable chain of NABL accredited calibration laboratory.		
Sr. No	Standard / Equipment	Id. No.	Cert. No.	NABL Cert. No.	Validity
1.	Tape & Scale Unit	QS/ME/75	SCL/23-24/11/204-001 dt 18/11/2023	CC-2282	17/11/2024

Observations : **Pre Calibration Status** : Worth calibrating,
: **Post Calibration Status** : As follows reading average value of three readings,

Mechanical Calibration: Dimension (Basic Measuring Instruments, Gauges etc.)

Parameter distance between(in mtr.)	Readings taken at master values (in mm)	Values of UUC (in mm)	Error (in mm)
0-1.0	999.955	1000.145	0.190
0-2.0	1999.955	2000.060	0.105
0-3.0	2999.955	2999.920	-0.035
0-4.0	3999.955	3999.865	-0.090
0-5.0	4999.955	4999.605	-0.350

NOTE

Maximum permissible error for total length in steel tape measure as per IS 1269(Part 2) is as below.(where 'L' is in meter)

Class I	Class II	Class III
± (0.1+0.1 x L)	± (0.3+0.2 x L)	± (0.6+0.4 x L)

1. For steel tapes of length upto 5 mtr. Permissible values are as per class I & II only.
2. For steel tapes of length above 5 mtr. Permissible values are as per class I, II and III.
3. For tapes in service the permissible value is equal to TWICE the values indicated in the above table.

Uncertainty of Measurement : ± 980 µm
(The reported uncertainty at above reading is the expanded uncertainty of measurement obtained by multiplying the standard uncertainty by the coverage factor k=2 corresponds to a confidence level of 95.45 %.)

Note: *UUC =Unit Under Calibration

END

Calibrated By — Ankita Bhuvad (Technical Assistant)

Reviewed 14/12/23

Approved By — N. M. Shinde, Technical Manager

User discretion:
Note: 1) Any error / Complaint must be resolved with lab within 3 days from the receipt of certificate in written.

Regd. Office & Labs : "Paras Complex", GP/67, Office No. 2 & 7 to 10 (Basement), Thermax Chowk, Chinchwad, Pune - 411 019.
Customer Care : 020-27371831, 07276710888 & Accounts : 020-27370204, Suggestion, Feedback & Complaint : 9822069138, E-mail : qrics@hotmail.com, web : www.qrics.in

P.T.O.